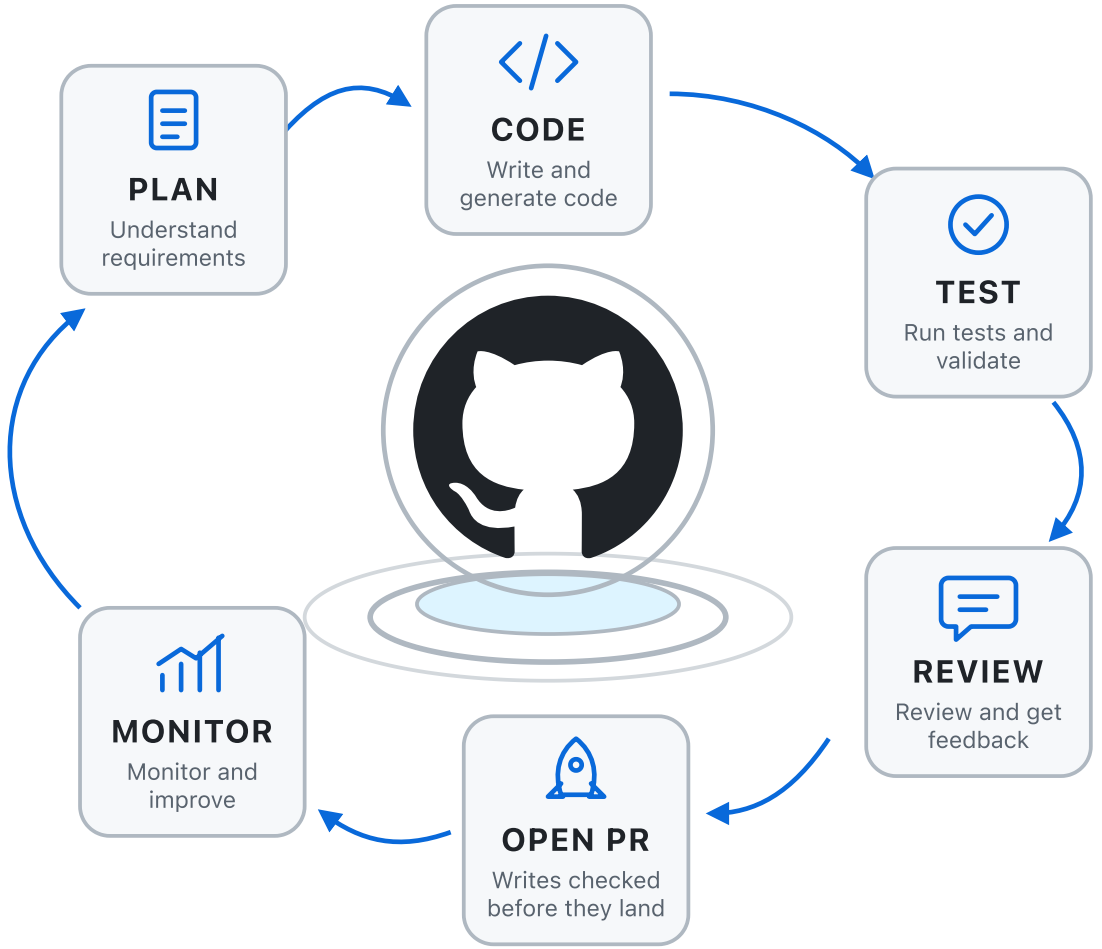
GitHub

Repos that keep themselves up to date

GitHub Agentic Workflows: from AI that suggests to AI that does

Donal Moloney · TODO: talk date and audience



Somewhere in our docs, a page is wrong right now, and nobody knows which one

56.5%

of 12,374 developers say **outdated information** gets in the way of the context they need at work.

Measured: Stack Overflow Developer Survey 2026. The page-is-wrong framing is this deck's; the survey measures the blocker, not our repositories.

Every rung on this ladder ships more code, and none of it describes itself



- **43.2 million** pull requests merged per month, up 23% year over year.
- **Nearly 1 billion** commits pushed in 2025, up 25.1%.

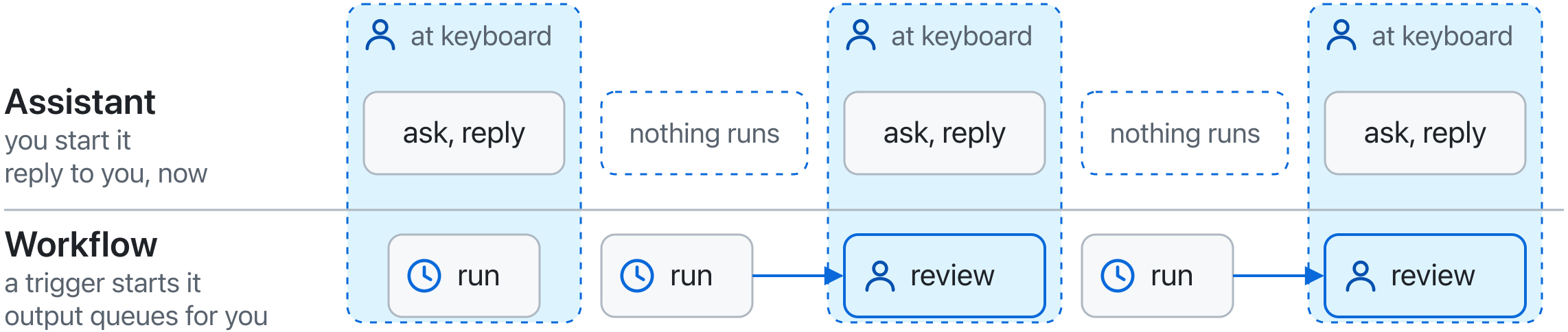
Measured: Octoverse 2025. Ladder: Feng et al., arXiv 2506.12469 — illustrative; L4 and the volume-to-docs link are this deck's reading.

I built something similar before, and it showed me what was missing

What I built	What worked	What it could never do
TODO	TODO	TODO

TODO: add a screenshot, and land the third column — it is the gap the rest of this talk fills.

An assistant waits to be asked; this one works a shift

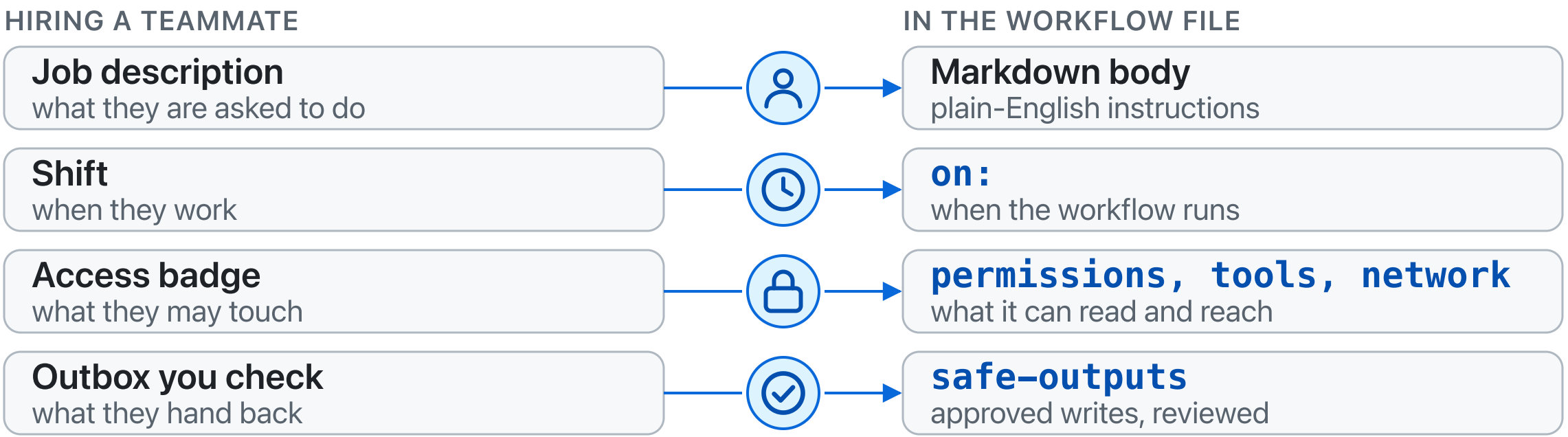


ILLUSTRATIVE · ONE WEEK, NOT TO SCALE · THIS DECK'S FRAMING

The chore nobody schedules is the chore that never gets done. That wrong page is exactly that chore.

Illustrative, and this deck's own framing. The only difference drawn here is who starts the work and where the output lands.

An agentic workflow is a GitHub Action you brief instead of script



Classic Action: you write the steps.

Agentic workflow: you write the brief, the agent plans the steps.

The teammate mapping is this deck's model, not GitHub terminology.

It does three kinds of work, and documentation is the safest corner to start in

MAINTAIN

Documentation maintenance · Issue triage · Dependency analysis

IMPROVE CODE

Pull request review · Security review

REPORT AND PLAN

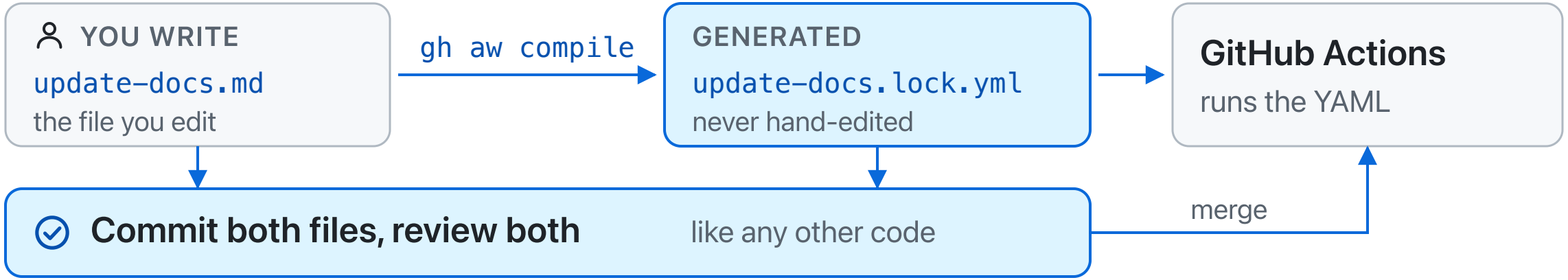
Metrics and analytics

We are asking for one job, in the first box: a single documentation page. It is the corner where a wrong answer costs a review comment, not an outage.

Groups are the gh-aw home page's own; the example tasks are from its gallery, verified 17 Oct 2026. Sorting those tasks into those groups is this deck's.

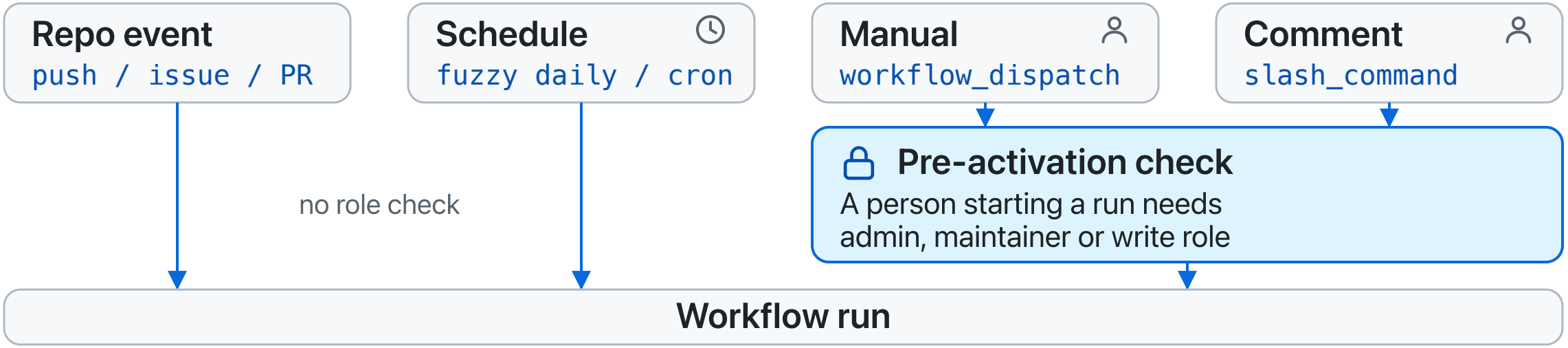
The instructions are plain English, and one command locks them into a GitHub Action

```
---
# on:, permissions, network, safe-outputs — the full file is next
---
Update the docs to match the code that changed, and open a draft PR.
```



Body paraphrased; the real file's instructions are longer.

Four things can start it, and a person who starts it must already have write access



Two ways to try it without touching our repository

- `gh aw trial` runs a workflow in a temporary private repository (experimental).
- `staged: true` under `safe-outputs:` runs it for real and writes nothing.

Thirteen lines say when it runs, what it can read, and the one thing it can send

```yaml
on:
  push:
    branches: [main]
permissions: read-all
network: defaults
safe-outputs:
  create-pull-request:
    draft: true
tools:
  github:
    toolsets: [all]
  web-fetch:
  bash: true
```

- **Shift:** every push to `main`
- **Access badge:** `read-all` — read-only; network limited to basic infrastructure
- **Outbox:** one draft pull request, and nothing else
- **Tools:** broad. `bash: true` runs arbitrary commands — in a job with no write permissions, behind a firewall.

The agent is granted no write access here. A separate job opens that one pull request, after a scan.

Frontmatter trimmed from `workflows/update-docs.md` in githubnext/agentics.

It will get things wrong, so it is built to be checked, not trusted

You would review a new teammate's first pull request. This one cannot skip that step.

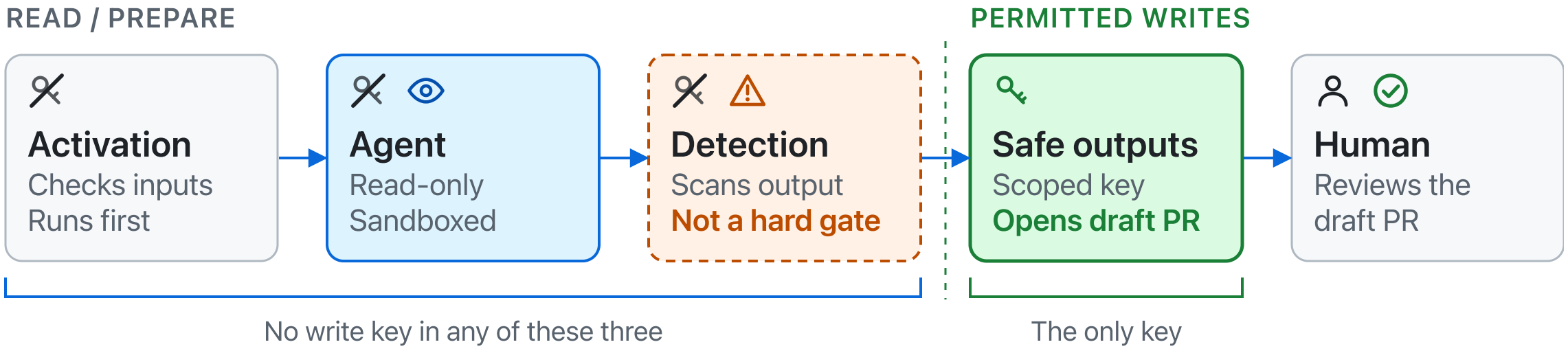
Wrong edits → it can only open a **draft** pull request, and a human reviews it. `draft: true` is a policy, not a default: the agent cannot override it. Never merged automatically: GitHub Blog guidance, not a hard limit.

Not a fit → it does not replace build, test or release pipelines. It is for work that needs interpretation, and nothing else.

Prompt injection → the agent is read-only and fenced in. The next two slides show how.

Cost → capped per run: 1,000 AI Credits (~$10) by default, and you choose the model. TODO: our own measured cost per run. gh-aw docs (safe outputs, cost management); GitHub Blog.

The agent never holds a write token: a separate, checked job does the writing

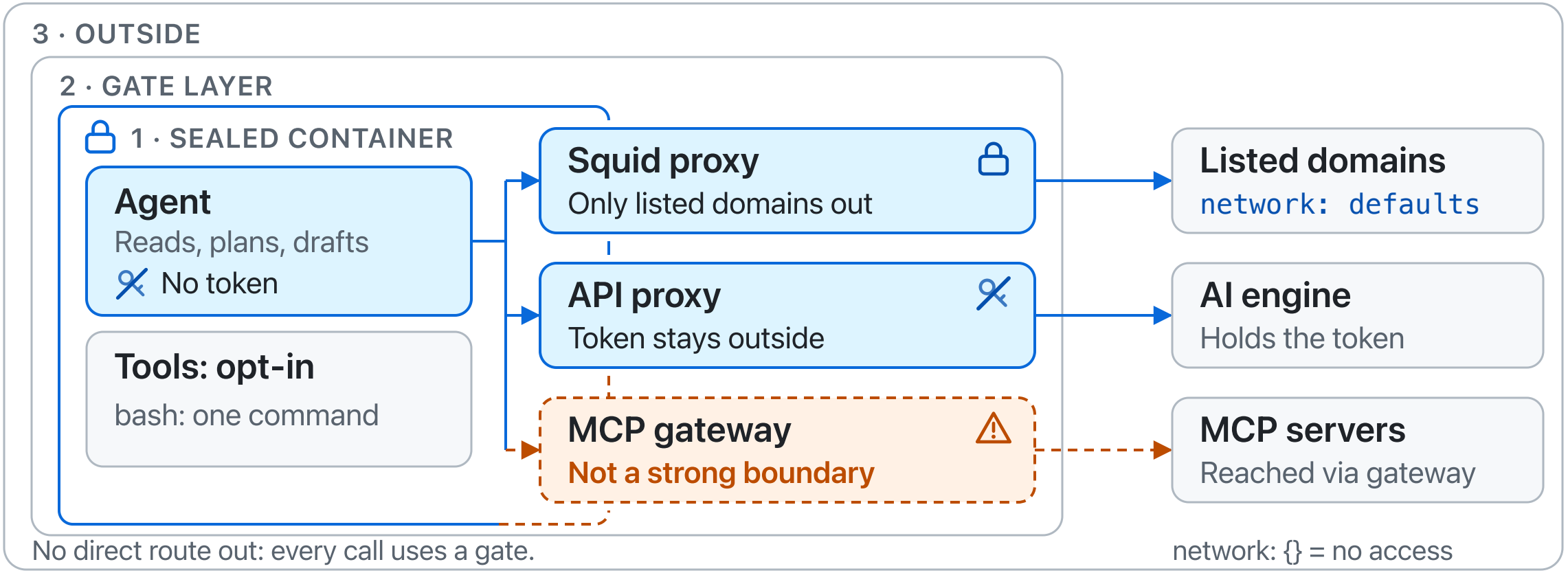


The teammate drafts the letter; someone else holds the key to the postbox.

- Detection scans every proposed output before anything is written. A failed check defaults to a caution notice, not a hard block (`continue-on-error` is true).

gh-aw architecture docs; four of its six jobs shown, plus the human review.

A misled agent is boxed in by layers, not by any single wall



gh-aw architecture and network reference. The MCP gateway caveat is the documentation's own wording.

GitHub reports that 57 of 59 pull requests from its docs workflow were merged

57 / 59

A 96% merge rate for its Daily Documentation Updater. Reported by GitHub: gh-aw blog, 13 Jan 2026; small sample. Merged is not the same as correct.

TODO: screenshots and recordings of a real pull request, issue and comment (link and length for each).

Ninety-eight of these are written and documented, and not one has ever run

Already written	Still unproven
98 workflow briefs in `workflows/`, one file each	**0** have ever run inside GitHub Actions
98 matching docs pages, one per workflow	**4 of 98** compiled locally; the other 94 are briefs
6 shared policy and prompt modules, imported rather than copied	No release tagged, and the sandbox repo does not exist

`DonalMoloney/Agentic-Workflows-Lib`, private. Measured: file counts via the GitHub API, 17 Oct 2026. Self-reported: the compile result and "not yet run" column, from the repository's own `agents.md`, not a CI result.

TODO: the one thing building this taught me, and whether the repo goes public before the talk.

Each of the 98 ships with a one-page spec: what it may write, and who owns it

	`readme-auto-updater`
Purpose	Spot drift between the README and the code after a push to `main`, and propose a fix
Shift	`push` — branches: `main`
Access badge	`permissions: contents: read`; GitHub toolsets `context`, `repos`
Outbox	**one** pull request — `create-pull-request`, `max: 1`
Owner	TODO: the real owning team
Maturity	**Experimental** — the label the library gives all 98

Trimmed from `docs/workflows/readme-auto-updater.md` and its frontmatter, verified 17 Oct 2026. Owner and install lines use a placeholder `acme-org` namespace and do not resolve.

Pick one stale page, and let an agent keep it current for a month

What I'm asking for today: TODO: one repository, one owner, one budget.

Week zero — find the page. The one whose code changed after the page last did. That's the wrong page from the opening, found rather than guessed at.

Weeks one to four — run it. `staged: true` first: it runs for real and writes nothing. Then draft pull requests only, every one reviewed. Staging previews outputs; it is not a sandbox (gh-aw docs).

How we judge it. Not whether we merged it, but whether it was **right** — the share a reviewer accepts as accurate. Merged is not the same as correct.

Then we decide. TODO: the result that means keep, the result that means stop.

You debug it like any Action: every run leaves logs you can audit

1. **Review the change:** the `.md` workflow and generated `.lock.yml` in the pull request.
2. **Watch the run:** the Activation, Agent, Detection and Safe outputs jobs — four of the six.
3. **Dig in:** `gh aw logs <workflow>` across runs, `gh aw audit <run-id>` for one run.

Try it yourself: my examples and the official docs

- TODO: my sites and example repository URLs
- `https://github.github.com/gh-aw` — the docs, and `/gallery` for ready-made workflows
- `https://github.com/githubnext/agentics` — where this deck's `update-docs.md` comes from
- `https://github.blog/changelog/2026-06-11-github-agentic-workflows-is-now-in-public-preview`

Status and sources

- Fact-checked 17 Oct 2026 against the gh-aw documentation, GitHub changelog, Stack Overflow Developer Survey 2026, Octoverse 2025, GitHub Next, and the GitHub Blog sources cited on each slide.
- Status: **public preview since 11 Jun 2026** (GitHub changelog). The gh-aw docs site still links a technical-preview deck; treat that as a leftover.
- Left off on purpose: the daily AI Credit cap (cost-management page says 5,000 per workflow per day by default; frontmatter page says `max-daily-ai-credits` is disabled by default), and "always blocks" for detection (`continue-on-error` defaults to `true`, turning detection warnings/failures into a caution notice).
- This deck's own models, not GitHub's: the teammate mapping, the assistant-versus-shift comparison, and placing gh-aw at L4 on the autonomy ladder.